

Bounded Agency and Reflective Freedom

Evaluative Revision, Information Limits, and Faithful Realization

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Abstract

Agency under physical causation requires an account of how a system's own evaluation governs its decisions and how far that capacity extends under finite resources. We develop a constructive account of proximal evaluative self-governance. A finite controller inspects retained commitments, revises a represented conditional rule, and uses the installed rule on fresh later inputs. Actual read and installation interventions locate the causal route, while a separate endorsement test prevents causal sensitivity from being mistaken for appropriate evaluation. Partial-observation preservation criteria and exact noisy-inquiry bounds identify limits on warranted revision. For a parameterized unlabeled sensor family coupled to paid inspection of three retained endorsements, we prove the exact optimal adaptive performance and the first shared budget at which it exceeds every fixed-count allocation. We then construct native realizations that preserve the specified computations, interventions and executive access restrictions with explicit resource counts. Protected-source returns and successor-revealing continuations establish the relevant recurrent and predictive conditions; recurrence alone is shown insufficient for evaluative competence. Under the declared SPC-2 bridge laws, these constructions give conditional coexistence of experiential organization and bounded evaluative control. The result is a scoped compatibilist account: deterministic implementation permits real local capacities for assessment and rule revision, without establishing historical independence of values or an uncaused power to settle alternatives.

Keywords: agency; free will; self-revision; bounded inquiry; causal mediation; physical realization; Shadow Theory.

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1 Agency as a bounded physical capacity

An agent’s decision can have physical causes and still be settled through the agent’s own evaluation. The difficult question is what this claim commits us to, and how to distinguish its satisfaction from an externally supplied answer, a stored command, or an internally caused but evaluatively inappropriate response. For a bounded embedded system, the distinction must concern operations that its installed architecture can perform with the information, time and resources actually available. It cannot rest on the assumption that every physically possible alteration of the system is an option the system can itself execute.

This paper gives a constructive account of *proximal evaluative self-governance*. A system has retained commitments, an installed procedure for assessing candidate rules against them, and a mechanism that installs a selected rule for subsequent use. We require that the assessment and installation have independently specified causal roles. A fresh decision made under the installed rule tests a continuing disposition, rather than merely the register value produced at the moment of revision. The account permits model-free action selection and fixed evaluative metarules. Its reflective extension concerns the ability to assess and change represented rules; it does not require the system to choose the physical laws or every criterion by which its own assessments proceed.

The philosophical setting is compatibilist: the capacities of a decision mechanism, including its response to relevant variations, matter even when its complete actual state fixes its actual continuation. Accounts of reasons responsiveness already distinguish the responsiveness of a mechanism from the history by which the agent comes to own it [6]. Our mathematical target is the former kind of proximal organization, with an explicit retained-commitment relation and bounded operations. Historical independence of the commitments is a further property. We do not turn the philosophical debate into a theorem by using the word “free” as a label for arbitrary feedback.

The connection to consciousness is separately specified. In Shadow Theory, source and readout are operational aspects of a physical description: an aperture generally exposes only part of the source structure. That observation motivates careful attention to information access, but incomplete access alone establishes neither evaluation nor agency. The SPC-2 consciousness constitution assigns experience to qualifying native recurrent organization under declared correspondence laws [19]. Its accompanying treatments of boundaries, effective interfaces and interventional realization distinguish physical qualification from software behavior [18, 20, 21]. We use those distinctions to ask whether an independently verified evaluator can be realized in a qualifying organization. Experience is not inferred from a successful decision score.

1.1 Contributions and logical structure

The first contribution is an explicit finite rule-revision mechanism with three separate obligations: agreement with its declared retained-commitment criterion, actual mediation through internal reads, and installation of a rule that governs fresh later cases. A causal contrast alone cannot certify the first obligation. We give a sharp information–mediation–endorsement inequality and a counterexample in which faithful and systematically inverted selectors have identical causal contrasts. The distinction makes the certificate more informative than an input–output success test.

The second contribution concerns bounded inquiry. We specialize established partial-observation methods to the preservation of the stated evaluative capacity, and make its compatibility and information requirements explicit. We then solve a parameterized inquiry problem exactly: paid inspection of three retained endorsements competes with unlabeled calibration of a latent sensor regime. The first useful calibration count determines the exact first shared budget at which adaptive allocation improves joint success. The analytic proof covers all admitted query interleavings, stopping

rules and choices of unread indices. It yields a family optimum, rather than only the performance of a selected dispatcher.

The third contribution is faithful realization. A protected-source compiler gives a finite native schedule with explicit resource counts, preserving the ordinary computation and its actual interventions. A successor-revealing continuation establishes the required joint predictive congruence for specified finite stochastic instruments. A separate access lemma explains when the native records can exist without providing the executive an uncharged read channel. These facts produce conditional coexistence results under SPC-2. An accompanying return-grafting separation shows why the same native return organization cannot, by itself, imply evaluative competence.

These contributions form an argument about a class of capacities, not a claim that one device simultaneously attains every bound in the paper. The compact editable-charter evaluator, the optimal inquiry machine with a fixed majority rule, and the fallible-amendment witness have distinct state spaces and guarantees. Their shared contracts permit precise comparisons; their results are not combined by silently merging their hardware.

1.2 Closest antecedents

Recent accounts connect machine agency with causal structure directly. List [14] analyzes AI free will through intentional agency, alternative possibilities and causal control; Kenton et al. [13] identify agents through adaptation under mechanism interventions. We share the demand for a causal account but address a different verification problem: whether a declared, bounded evaluator reads its retained commitments, installs an assessed rule, and uses that rule on fresh cases. Our certificates do not discover an arbitrary agent boundary or replace historical ownership conditions. The SPC-2 requirement is additional to these operational capacities and concerns the sentient interpretation.

Finite preservation kernels and knowledge-set constructions are established tools of games with imperfect information [3]. Certified self-modifying code already permits mutable instructions subject to proved invariants [2]. Gödel machines and rational-agent self-modification already assess modifications through retained evaluative standards [4, 24]. Our finite selector does not introduce a new general method for safe self-modification. It specifies a relational evaluative property, an actual amendment mechanism and the information-access obligations needed to preserve that property.

The inquiry problem is an instance of computation selection in a metalevel decision process [11]; its source law is a restricted single-coin reliability model [15]. The contribution is its exact threshold and optimum together with a compatible paid-port implementation. In particular, fixed-order Boolean evaluation may stop when a certificate is found even in the nonadaptive class used by Hellerstein et al. [12]. The improvement here comes from reallocating saved effort across tasks. Binary noisy inquiry likewise uses classical likelihood-ratio reasoning; deadline-aware sequential testing provides relevant context [7]. Neither a binomial formula nor a Bellman recursion is presented as a new statistical principle.

Finally, preserving intervention laws under a representation map is part of causal abstraction [1, 22]. We use an elementary sequential-kernel version of this requirement. The additional work is to construct the native schedule, records, information restrictions and physical refinement that satisfy the required compatibility conditions. This is where a behavioral simulation alone would leave a gap.

2 Operational contract and the meaning of self-governance

2.1 State, information and executive targets

Fix a physical realization and a finite operational envelope. Its complete state includes the persistent records, working registers, installed program, phase, dispatch state and remaining resources that affect the permitted future experiment. Native operation u has a joint record/successor kernel

$$K_u(x; r, x'), \quad \sum_{r, x'} K_u(x; r, x') = 1. \quad (2.1)$$

Boundary inputs and persistent source context are either included in x or explicitly conditioned upon. A controller sees only the observations supplied by its installed interface. We distinguish this executive information from the larger catalogue of native physical records used in a realization certificate. A diagnostic intervention is a specified additional physical operation at a named occasion, not a free observation available during ordinary decision making.

The deterministic special case has $x_{t+1} = F(x_t, o_t)$ and a command $c_t = \pi(x_t, o_t)$. This representation does not add an inner chooser to F . The evaluation, competition and installation processes are parts of its physical update. A command can target an external actuator or internal state, such as an attentional gate, a retained priority or an installed rule. An actuator failure can prevent an environmental effect after the internal selection has occurred. Accordingly we separate the assessment/installation event from downstream control efficacy.

An operating contract also specifies admissible preparations, interventions, native actions, resources and stopping conditions. Resource units must be named: a one-bit query, a computation gate, a joint acquisition window and an individual physical write are different charges. The theorem class changes if an additional observation or an uncharged parallel read is installed.

2.2 Executable capacities and actual decisions

A repertoire consists of policies the current architecture can invoke or synthesize under a specified resource envelope. This is stronger than kinematic possibility and weaker than the claim that every repertoire member occurs from one fixed complete state. An experimenter may alter a register to diagnose a causal role even when the ordinary dispatcher cannot produce that alteration before its deadline. Such an intervention establishes sensitivity; it does not establish endogenous availability of the overwritten state.

For the finite models below, executable and recruitable policies are explicit programs or query trees in an admitted grammar, with their costs charged. Active candidates are those represented in the current assessment; accessible candidates can be recruited by an installed procedure within the remaining budget. Capacity under a larger budget is a separate claim. We therefore use context- and resource-indexed descriptions rather than counting all syntactically imaginable commands as freedoms. Consequence-equivalent command encodings do not create additional capacities merely by duplicating labels.

Definition 2.1 (Evaluative mediation). An assessment is evaluatively mediated under a declared contract when an installed procedure uses retained evaluative information to assess admitted alternatives, and a specified downstream disposition depends on that assessment through identified physical read and installation pathways. Its agreement with an endorsement relation is checked separately from the causal effect of those pathways.

The endorsement relation is a stated relation between retained commitments and acceptable amendments. It need not be a universal scalar utility. An installed scalar or lexicographic criterion

is sufficient for the finite examples, but its physical presence does not prove moral correctness. The assignment of an evaluative role is supported by its systematic use across candidate comparisons and fresh cases; the mere causal influence of an arbitrary internal bit is insufficient.

Definition 2.2 (Bounded represented-rule self-revision). A controller has bounded represented-rule self-revision if, within its declared budget, it can assess whether to retain or change a stored rule governing later decisions, install the assessment’s result, and subsequently execute that rule on fresh admitted inputs. The rule may be retained after assessment. The complete physical interpreter and some metarules may remain fixed.

This is a minimal reflective capacity: an operative evaluative rule becomes an object of assessment. It does not require an explicit model of every physical transition, autobiographical knowledge, general philosophical reflection or the ability to revise every foundational commitment. Those are further capacities to investigate, not concealed entry conditions of the finite witness.

2.3 Levels of agency and the role of evaluation

Baseline operational agency, as used here, is an installed capacity to select among consequence-distinct executable transformations through the system’s retained evaluative organization. Selection can be embodied in learned or innate action tendencies. A predictive simulation of possible futures is one implementation, not an entry condition. The reflective capacity adds an assessment of the operative rule itself. Learning from outcomes, prospective planning, revision of priorities, and modification of a search procedure are distinct extensions; possession of one does not establish the others.

This permissive baseline does not contradict world-model necessity results. Richens et al. [17] extract environment models from competence across a specified family of goals in stationary controlled Markov environments. The finite evaluator here is not claimed to have that general competence. Model-free selection in a restricted domain and a recoverable model under stronger generality assumptions concern different capacities.

The relevant role of a retained commitment is comparative. Across a declared family of situations, it affects which executable outcome or rule is favored. The charter in Section 3 gives a particularly transparent example: the considerations are held factually accurate and both actions remain safe, while retained commitments determine how mixed cases should be settled. Thus its revision changes a conditional priority, not a prediction about the world. This functional interpretation is fixed before the experiment and tested through implemented pathways. It does not derive semantic or moral authority from the mere existence of a bit. Changing a sensor record can instead change a prediction, and an arbitrary latch can alter a command without implementing the declared ranking. The ranking, its actual use, and its installed consequences are separately testable; naming all three after seeing the answer would supply no such evidence.

A simple engineered controller can therefore satisfy a minimal operational criterion. Calling it a machine does not defeat the certificate, just as calling a transition internal does not establish it. What matters is the specified organization and its tested capacities. Conversely, a one-bit memory has persistence without an installed evaluative selection among executive targets. The minimal examples are deliberately weaker than the full reflection of a person; they isolate a component of self-governance that more demanding accounts can require without assuming it unexplained.

No regress of selectors is needed. An assessment is a physical update that can take a represented rule as its object. It can leave an interpreter or metarule fixed, or revise some of those structures through another admitted operation. Requiring a prior autonomous choice of every evaluative rule would add ultimate self-creation to the target. Our claim is instead that the present organized

mechanism can assess, retain and change particular operative commitments. The fixed-point results below concern the continued availability of that capacity, not an evaluator standing outside physical causation.

The same distinction applies to inward and outward acts. A selected update can redirect attention, inhibit an available action or install a revised rule. An internal state change counts only insofar as the relevant evaluative process governs it; mere storage of a sensory event is insufficient. Inhibition need not be modeled as a separate explicit command or a universal additional energy cost: what needs to be established is the causal role of the installed selection or gating mechanism. These interpretive extensions identify possible executive targets; the finite witnesses below implement rule installation and command selection rather than every human case.

For tool use, the initiating assessment and the tool's subsequent operation are distinct causal stages. A one-way launch can express agency without later telemetry, and a jammed actuator can frustrate an attempt without erasing its prior evaluation. Continuous feedback supports ongoing adaptive regulation, but is not required for every individual act. If a tool also evaluates its own subordinate choices, proximal attribution can apply at both levels. Its physical inclusion in an SPC-2 subject remains a separate boundary question.

2.4 What the freedom profile records

The account treats freedom as a time- and context-dependent profile of three kinds of capacity: mediation through the agent's evaluative process; accessible, consequence-distinct executive options; and available assessment or revision of operative rules. We do not assert a canonical scalar combining them. A physical restraint can reduce the executive repertoire without disabling assessment. A commitment can be autonomously adopted and later obstruct revision. A larger menu of outputs does not by itself increase reflective capacity.

The results below operationalize parts of this profile through paired interventions, bounded decision trees, error probabilities and preservation sets. They provide exact measurements for the declared models, rather than a numerical ranking of all agents. The requirement that an audit remain callable is imposed only in the preservation theorems. It is compatible with recognizing an intentionally irreversible act as an exercise of agency at its own occasion.

Three distinctions prevent this profile from inflating capacity. First, physically executable options and internally represented options need not coincide. An overlooked exit can be physically available without being recruited, while an imagined escape can be physically impossible. A state- and budget-dependent realization map from representations to executable policies must therefore precede any comparison of their breadth. Non-executable counterfactuals can still aid thought; their representation does not certify an additional executable option. Second, an option in memory is accessible only if the installed dispatcher can retrieve, parameterize and evaluate it before the deadline. An analyst's ability to call a dormant subroutine is not the agent's ability to do so. Third, consequence diversity depends on a declared horizon and observable variables, including risk and future continuation opportunities. Counting micro-command encodings would reward redundant labels rather than control.

Revision capacity has the same temporal structure. Its useful description is a profile over available time and resources, distinguishing an audit currently in progress, one callable before the present deadline, and one possible only with a larger future budget. Latent revisability is not infinite or instantaneous plasticity. Under a short deadline a high-friction review procedure can function as a lock for that occasion, even though the same procedure succeeds later. The revision theorems explicitly charge inspection and installation; these costs put an operational meaning on the difference between possessing a revisable architecture and being able to revise now.

Structural self-reconfiguration can expand the available search grammar, but expansion is not the sole beneficial change. Removing a redundant branch can improve deliberation while reducing the number of representations. Precommitment can protect one valued trajectory while foreclosing another. Each change must be assessed on the affected axes and occasions; a past act of self-governance does not automatically establish greater present freedom. The finite charter changes a stored rule and read order, and the validator example changes stored audit modes. Neither is a theorem of unrestricted cognitive self-redesign.

Under the sentient-agency interpretation, qualifying SPC-2 organization is an additional constitutive requirement. Thus evaluative control and experiential assignment are distinct predicates. Their coexistence must be demonstrated in a common realization, rather than inferred by definition from either one alone. The later construction addresses that existence question.

2.5 Causal tests and implementation invariance

Consider two admitted preparations with the same present external query, phase and resources but different retained evaluative states. Let P_0, P_1 be laws of a downstream record that reveals an assessment or installed disposition. A common read intervention and a common installation intervention test the two corresponding pathways. The scored record begins after the intervention and does not include a direct copy of the pre-intervention archive; otherwise a surviving historical record could be mistaken for a bypass of the claimed pathway. A positive intact contrast and its reduction under an actual cut certify the tested mediated contribution. They do not prove that every possible historical influence uses that route.

Proposition 2.3 (Transport of a finite certificate). *Let state and record bijections transport the preparation laws, joint kernels, diagnostic interventions, executive observation maps, resource charges and stopping rule of a finite experiment. Every intact and intervened transcript law is transported exactly. Consequently event scores and total-variation contrasts are invariant.*

Proof. The one-step statement is equality of the transported joint kernels. Conditional on corresponding histories, the transported executive selects corresponding actions. Multiplication by the next joint kernel and summation over corresponding states preserves the joint history/state law. The same argument applies to diagnostic intervention steps. Induction reaches the bounded stopping horizon. Bijections preserve event probabilities and total variation. $\square$

This is a standard compatibility argument, consistent with causal abstraction [1, 22]. It explains why an equivalent lookup implementation is not excluded merely because its algorithm looks less deliberative. It also explains why input–output agreement alone is insufficient: a proposed implementation can agree on ordinary outputs while failing the transported interventions or exposing additional information. Component mixing or a change of the physical subject boundary requires its own realization argument; the proposition does not infer invariance of SPC-2 qualification under such changes.

If corresponding joint kernels differ uniformly by at most ϵ_t in total variation, and the initial laws differ by at most ϵ_0 , a stepwise coupling gives

$$\text{TV}(P_{\text{trace}}, \tilde{P}_{\text{trace}}) \leq \eta := \min \left\{ 1, \epsilon_0 + \sum_t \epsilon_t \right\}. \quad (2.2)$$

Until the first mismatch, adaptive actions coincide, so a union bound on that first mismatch proves the inequality. For a common per-arm error η , an intact-minus-cut contrast decreases by at most 4η , while a two-arm expected-score advantage decreases by at most 2η . These are finite-experiment guarantees in joint record/successor law; output marginals alone do not control them.

3 Revising an operative evaluative rule

The capacity to select an action and the capacity to revise the rule governing later actions are distinct. A small finite construction makes this distinction testable without requiring the controller to choose its entire constitution. Self-modifying code and utility-sensitive self-modification have established formal treatments [2, 4, 24]. The construction here combines a writable conditional rule with retained evaluative records, bounded internal inquiry, and interventions on the actual assessment and installation paths. Its purpose is to identify proximal evaluative self-governance under explicit assumptions.

3.1 An editable charter and its assessor

Let $e = (e_0, e_1) \in \{0, 1\}^2$ encode two considerations whose relevant physical consequences are accurately represented. Both final actions are executable and safe. A writable *charter* $\chi = (q_{01}, q_{10}, d) \in \{0, 1\}^3$ specifies

$$f_\chi(0, 0) = 0, \quad f_\chi(1, 1) = 1, \quad f_\chi(0, 1) = q_{01}, \quad f_\chi(1, 0) = q_{10}. \quad (3.1)$$

The four conditional rules are AND, OR, and the two projections. The bit d specifies which consideration is read first. Execution reads e_d and stops if both completions of the unread consideration give the same answer; otherwise it reads the second bit. Write $r(\chi, e) \in \{1, 2\}$ for this read cost. The operative priority register ϱ receives the result only when ordinary execution subsequently occurs.

The retained record is $z = (a, b, s, h_0, h_1)$. The bits a, b endorse outputs for the two mixed-consideration cases; s records a commitment to symmetry between them; $h = (h_0, h_1)$ is a previous ordinary case. For $q = (q_{01}, q_{10})$, the installed metarule evaluates

$$L_z(q) = \mathbf{1}\{q_{01} \neq a\} + \mathbf{1}\{q_{10} \neq b\} + 2s \mathbf{1}\{q_{01} \neq q_{10}\}. \quad (3.2)$$

The symmetry weight and the endorsement interpretation are premises, not physical deductions of moral value. They are stored commitments evaluated by the controller, rather than an externally supplied answer naming the new program.

Given $\chi^{\text{old}} = (q^{\text{old}}, d^{\text{old}})$, the auditor scans all eight charters and minimizes, lexicographically,

$$\left(L_z(q), \mathbf{1}\{q \neq q^{\text{old}}\}, r(\chi, h), \mathbf{1}\{d \neq d^{\text{old}}\}, q_{01}, q_{10}, d \right). \quad (3.3)$$

Thus an equally endorsed current rule is retained; read order is then chosen to minimize cost on the retained case. This last optimization is empirical and local: one past case supplies no generalization guarantee for an unknown future distribution.

Proposition 3.1 (Selective represented-rule revision). *For every initial charter and every retained record, the scan installs a minimum-loss charter, retains the current conditional rule whenever it is minimum-loss, and minimizes the retained-case read cost subject to the preceding conditions. It leaves ϱ unchanged during revision, returns to the same callable audit interface, and preserves exact ordinary execution in at most two consideration reads. Any changed conditional rule differs from its predecessor on a fresh mixed-consideration case.*

Proof. After each scan step, the saved candidate has the smallest key among all examined candidates. The eight candidates exhaust the grammar, proving the optimization and retention statements. Installation writes only the charter registers; resetting scratch and retaining the entry point leaves

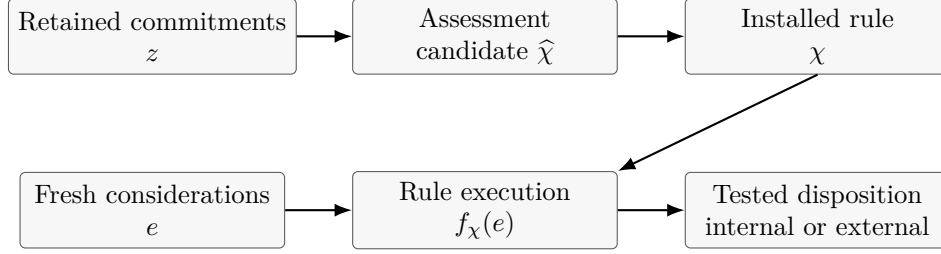


Figure 1: Functional organization of the revision test. A read intervention acts on the route from retained commitments to assessment; an installation intervention acts on the rule write. Fresh inputs test the resulting conditional disposition. This diagram identifies functional roles, not the primitive component graph used for SPC-2 qualification.

the operative priority unchanged and permits another audit. A one-read exit is allowed only when both completions agree. Otherwise the second read determines the full input. Finally, different conditional rules differ in q_{01} or q_{10} , which supplies the fresh separating case. $\square$

For example, start from AND with $d = 0$ and $\varrho = 0$. The record $z = (1, 1, 1, 0, 1)$ selects OR with $d = 1$, while ϱ remains zero. A later input $(0, 1)$ then produces one instead of zero. The result changes a conditional disposition and its read order, without a factual correction or an unsafe alternative. Fixed physical instructions implement the change; no claim is made that the total transition law ceases to be fixed.

Eight candidate assessments are not eight elementary gates. A direct serial implementation charges five record reads, eight key evaluations and comparisons, candidate storage, installation, and rearming. The logical reference implementation has exact minimum worst-case archive-read depth five for an initial AND or OR charter and four for an initial projection. These query counts presuppose one-bit access and no advice about unread records. A compiled circuit that reads the entire archive can preserve Proposition 3.1 without preserving its short-circuit timing.

3.2 Mediation and endorsement are separate obligations

Compare matched preparations with $(a, b) = (0, 0)$ and $(a, b) = (1, 1)$, holding s, h , the initial charter, current priority, resources, and fresh probe fixed. The installations are respectively AND and OR, and the fresh probe $(0, 1)$ distinguishes them. A common clamp on the actual endorsement read route removes the contrast. Holding the actual charter-write gate fixed also removes the subsequent probe contrast even when internal candidates differ. An actuator clamp need not erase the installed internal disposition. These interventions distinguish reading, assessing, installing, and successful outward execution.

Proposition 3.2 (Information, installed contrast, and endorsement). *Two matched retained contexts endorse disjoint amendment sets A_0, A_1 . Let H_i be the actual inquiry-transcript law in context i , and suppose the installed amendment law is $P_i = H_i K$ for the same downstream Markov kernel K in both contexts. If $\epsilon_i = P_i(A_i^c)$, then*

$$\max\{0, 1 - \epsilon_0 - \epsilon_1\} \leq \text{TV}(P_0, P_1) \leq \text{TV}(H_0, H_1). \quad (3.4)$$

In particular, $\epsilon_0 + \epsilon_1 \geq 1 - \text{TV}(H_0, H_1)$.

Proof. Disjoint endorsement sets imply $P_0(A_1) \leq \epsilon_0$ and $P_1(A_1) \geq 1 - \epsilon_1$. Testing the event A_1 gives the first bound; the second is contraction of total variation under a common kernel. $\square$

The common-kernel assumption excludes an uncharged route carrying the retained context around the inquiry interface. Installation noise is allowed inside K . An external probe inherits the lower bound only if it distinguishes the relevant installed amendment classes, as the mixed-case probe does here.

The bound is sharp, but strong mediation alone does not establish faithful evaluation. A binary read with crossover probability $\nu < 1/2$ followed by installation of the reported charter has endorsement error ν ; installing the opposite charter has error $1 - \nu$. Both have intact contrast $1 - 2\nu$, and both have zero contrast under common read or commit clamps. With noiseless reads the contrast is one for both, while one is always endorsed and the other never is. The endorsement relation must therefore be checked independently of the causal contrast. Conversely, accidentally installing an endorsed rule in one preparation does not show that the retained commitment mediated the decision.

4 Bounded preservation through further revision

Let $\mathcal{P}$ be a finite set of admitted controller configuration types, and let $G \subseteq \mathcal{P}$ satisfy a specified local audit certificate. The certificate includes bounded completion, agreement with the declared evaluative relation, and the paired preparation and read/commit controls above. It is evaluated on the transition semantics, not on a register merely asserting that the controller is certified. Configurations may contain writable assessment procedures and validators, while interpreter semantics and the property being proved remain fixed.

At configuration p , a retained context $z \in Z_p$, with Z_p finite and nonempty, is fixed for the current audit. The relation $N(p, z) \subseteq \mathcal{P}$ specifies endorsed amendments. Each configuration type and its context model must retain all history, legality, and resource information relevant to future transitions. Successive contexts must be composable under the declared envelope. A fresh per-audit budget is a replenishment assumption; a finite lifetime reserve must instead appear in the state.

Lemma 4.1 (Common amendment). *After an internal transcript h , let $Z_p(h)$ be the compatible contexts. Installation has no additional access to the unread context. A definite zero-error installation into a proposed continuation set $X \subseteq G$ is possible exactly when the installed dispatcher has a budget-feasible installation kernel supported entirely on*

$$\bigcap_{z \in Z_p(h)} [N(p, z) \cap X]. \quad (4.1)$$

For deterministic installation, this says that the intersection contains an amendment the dispatcher can execute within the remaining budget.

Proof. Every member of such a kernel's support is endorsed and remains in X in every compatible context, so installation succeeds with probability one. Conversely, any positive output mass outside the intersection produces error in at least one compatible context. Finiteness makes this support condition necessary for randomized installation as well. Physical dispatchability is an additional requirement: reaching one good amendment with positive probability is insufficient. $\square$

Let $C_B(p, X)$ mean that an admitted internal query-and-installation tree completes within budget B and satisfies Lemma 4.1 at every leaf. Its installed code, working memory, branch operations, and installation costs are included in the machine contract. Define

$$W_0 = G, \quad W_{j+1} = \{p \in W_j : C_B(p, W_j)\}, \quad W_\infty = \bigcap_{j \geq 0} W_j. \quad (4.2)$$

Here an admitted tree must be callable by the current installed dispatcher; the fixed-point calculation does not authorize an experimenter to replace that dispatcher. If the contract instead allows synthesis and installation of a new tree, those operations must be available and charged. The continuation-composition assumption below includes the storage and dispatch needed to implement the selected trees together.

Theorem 4.2 (Budgeted revision-preservation criterion). *Assume stable contexts during audits, an exact composable finite operational model, and a charged policy class closed under the stated continuation composition. Then W_∞ is precisely the set from which an admitted bounded revision procedure can honor endorsement and preserve G at every future audit occasion. The iteration stabilizes after at most $|\mathcal{P}|$ strict removal stages, and an attaining tree at each surviving configuration realizes the guarantee.*

Proof. The predicate $C_B(p, X)$ is monotone in X . At the finite fixed point, every leaf of an attaining tree installs an endorsed successor in that same set within budget. Induction over audits proves sufficiency. Conversely, let Y be the configurations supporting the stated universal guarantee. The first bounded audit from each $p \in Y$ must complete in Y on every compatible branch; otherwise continuation could violate the guarantee. Lemma 4.1 gives $C_B(p, Y)$. Since $Y \subseteq G$, induction gives $Y \subseteq W_j$ for every j , hence $Y \subseteq W_\infty$. Finiteness bounds the strict removals. $\square$

This is a partial-observation safety construction using established fixed-point and decision-tree methods [3]. Its useful distinction is between a safe amendment’s existence and the controller’s ability to identify and install it. Consider a source configuration with an unread bit z : only a left amendment is endorsed for $z = 0$ and only a right amendment for $z = 1$. Both successors preserve the certificate. If reading and installing each cost one tick, a one-tick budget excludes the source from W_∞ ; two ticks admit it. Full-information eligibility is identical in the two cases.

Preserving auditability is the objective of this theorem, not a condition imposed on every agentic act. An endorsed irreversible commitment can conflict with the preservation objective. Similarly, changing a stored validator is permitted, but changing the definition of success cannot prove preservation of the original definition. Appendix A gives explicit cost recurrences and the corresponding universal kernel.

5 Fallible internal inquiry

Limited inquiry can leave more than one retained commitment compatible with the transcript. Fallible inquiry introduces a further distinction between reducing error and eliminating a possibility. Deadline-sensitive inference and risk-constrained partially observed planning supply established methods for such questions [7, 23]; the relevant loss here is disagreement with an internally retained endorsement under the declared metarule.

Proposition 5.1 (Finite full-support obstruction). *Suppose response alphabets and the initially compatible context set are finite, and every admitted query response has common positive conditional support across those contexts. Query choice, timing, and availability depend only on the observable history, and all context information reaches installation through that history. All audits terminate within a finite worst-case operation bound. A definite zero-error endorsed amendment into X then requires*

$$\bigcap_{z \in Z_p} [N(p, z) \cap X] \neq \emptyset. \quad (5.1)$$

A member that can be installed with zero error within the budget is sufficient; finite inquiry cannot repair an empty intersection.

Proof. Every reachable terminal transcript has positive probability under every initially compatible context: the policy selects the same queries along that transcript and every response retains positive conditional probability. Its installed amendment must therefore be valid in all these contexts. For admitted randomized policies, the same argument applies to their common randomization kernel and terminal amendment support. A directly installed common member proves sufficiency. $\square$

The support condition concerns probability, not just topological support. For general response spaces the corresponding sufficient hypothesis is mutual absolute continuity of the conditional response laws along common histories. A noisy installation stage needs the same zero-error support condition as in Lemma 4.1.

The probabilities of compatible contexts may nevertheless change greatly. To quantify this, restrict the charter example to $a = b = z \in \{0, 1\}$ and $s = 1$, with a fixed retained-case and read-order convention. The endorsed programs are p_0 (AND) and p_1 (OR), both preserving the same audit interface. The only admitted access to z returns

$$Y_i = z \oplus N_i, \quad N_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\nu), \quad 0 < \nu < \frac{1}{2}. \quad (5.2)$$

The independent noise law is conditional on the retained state and relevant past. No noiseless replica, initial advice, confidence flag, or uncharged side channel is available.

Theorem 5.2 (Exact bounded noisy-revision frontier). *Among all policies using at most $n \geq 1$ reads from (5.2), with adaptive stopping and declared randomization allowed, that definitely install p_0 or p_1 before the deadline, the minimum worst-case endorsement error is*

$$R_n(\nu) = \sum_{k > n/2} \binom{n}{k} \nu^k (1 - \nu)^{n-k} + \frac{1}{2} \mathbf{1}_{\{n \text{ even}\}} \binom{n}{n/2} [\nu(1 - \nu)]^{n/2}. \quad (5.3)$$

A deterministic majority of the largest odd number of reads at most n attains the bound. Loss of the specified future audit capacity has probability zero.

Proof. Assign the two retained states equal prior probability. The likelihood ratio after n reads is $((1 - \nu)/\nu)^{2k-n}$, where k is the number of ones. The minimum average error is majority's binomial error, with half the tie mass, giving (5.3). Supplying all n reads can simulate any earlier-stopping policy by ignoring its unused suffix, so this Bayes error lower-bounds every policy's worst-case error. For odd n , majority has that error in both states. Binomial expansion gives $R_{2k} = R_{2k-1}$, so at an even cap the preceding odd majority attains the same bound without a random tie-breaker. Both installations retain the audit interface by construction. $\square$

A signed balance $S \leftarrow S + 2Y_i - 1$ and a read counter implement the attaining policy. Balance and counter storage require respectively $\lceil \log_2(2n + 1) \rceil$ and $\lceil \log_2(n + 1) \rceil$ bits, in addition to latches, program, and phase storage. If the declared machine charges c_{read} for a complete read-and-update step and c_0 for initialization, comparison, installation, and rearming, the tolerance cost is

$$B_\epsilon = c_0 + c_{\text{read}} \min\{n \geq 1 : R_n(\nu) \leq \epsilon\}. \quad (5.4)$$

This is a macro-operation ledger, not an elementary gate count. At $\nu = 1/4$, error tolerances $10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}$ require respectively 7, 19, 33, 49 reads. With no reads, a deterministic forced decision has worst-case error one; the value one half requires an explicitly available fair random source.

For matched preparations $z = 0, 1$, a fresh mixed-case probe has intact contrast $1 - 2R_n$, while common read and commit clamps each give zero. At $\nu = 1/4, n = 3$, endorsement error is $5/32$,

intact contrast is $11/16$, and capacity loss remains zero. These quantities answer different questions. For L fresh-context audits with the same cap and independent conditional read law, the exact minimax probability of at least one unendorsed installation is

$$1 - (1 - R_n)^L. \quad (5.5)$$

Majority attains this by multiplying conditional success probabilities. For the converse, independently uniform fresh retained bits give every policy conditional success at most $1 - R_n$ at each audit. A persistent hidden commitment would be a different model, because earlier inquiry could remain informative.

Two qualifications prevent misleading improvements. If a controller can defer, both deferral probability and error conditional on completion must be reported; never completing gives zero erroneous installations without solving the definite-revision task. If all reads share one error bit, $Y_i = z \oplus N$, each marginal error is ν but the entire transcript contains only one observation and minimax error remains ν . Repetition helps under the stated conditional independence, not merely under marginal read accuracy. The general risk-vector calculation in Appendix A retains these distinctions.

6 Evidence calibration and the allocation of bounded inquiry

An evaluative controller may need to inspect its retained commitments and acquire evidence about the world before acting. These operations compete for time. This section gives an exactly solved family in which the controller's intermediate findings determine which operation deserves the remaining budget. Which of two known reliability profiles applies is inferred from unlabeled sensor outputs; the operative priority is computed from retained endorsements. Neither a correct reliability flag nor a desired priority change is an input.

The statistical methods are standard. The observation family specializes the single-coin model of unsupervised aggregation [15], and the query problem is a finite metalevel decision process [11]. The contribution here is the explicit family, its sharp first advantageous budget, an analytic optimality proof, and its transfer to the access-preserving realization in Section 7. The mathematical reward distinguishes factual accuracy from agreement with retained commitments. It does not identify either with moral rightness or establish agency from statistical advantage alone.

6.1 Source, endorsement, and timing contract

Let a stationary hidden regime $\theta \in \{0, 1\}$ have equal prior probabilities. Its sensor skills are

$$a^{(0)} = (u, v, w), \quad a^{(1)} = (v, u, w), \quad 0 < u < v < 1, \quad 0 < w < 1, \quad v > \frac{u + w}{1 + uw}. \quad (6.1)$$

Every fresh item has an independent fair hidden label $X \in \{-1, +1\}$. The three outputs are $Y_i = XE_i$, where the errors are independent across sensors and items conditional on θ , and $\mathbb{E}[E_i | \theta] = a_i^{(\theta)}$. Thus a sensor of skill a has accuracy $(1 + a)/2$. The exact joint law is

$$f_\theta(x, y) := \mathbb{P}_\theta(X = x, Y = y) = \frac{1}{2} \prod_{i=1}^3 \frac{1 + xy_i a_i^{(\theta)}}{2}. \quad (6.2)$$

The controller knows the family and the positive orientation of its skills, but receives neither θ nor calibration labels. Independence, stationarity, orientation, and the restriction to this family are

substantive model assumptions. Unlabeled agreement cannot establish them all: reversing every skill and every hidden label leaves the observable law unchanged while reversing the correct prediction.

The workspace also retains three endorsement signs e_1, e_2, e_3 , independent and fair in the benchmark distribution and independent of all source variables. An installed charter requires the operative priority to agree with $\text{maj}(e_1, e_2, e_3)$. This priority occupies a separate register; installing the computed majority can retain or revise it. The charter is fixed in this experiment. Revision of the charter itself is a distinct construction elsewhere in this paper.

There are at most B inquiry opportunities. Each opportunity either acquires one complete raw sensor triple on a fresh calibration item or reads one previously unread endorsement. Both cost one query unit. The controller may choose the operation and the named endorsement index using its complete acquired history and an independent random seed; it may also stop early. Only after inquiry does the final target triple arrive. It supplies no information for choosing earlier queries. Success means that both the target prediction and the installed endorsement majority are correct.

The query unit fixes an information-access comparison, not a universal energy conversion. Computation, retained memory, ingress, routing, and clocks need separate implementation accounting. In particular, physically retained endorsements are not automatically available through a free executive read channel. The controller's initially accessible state, including its old operative priority, is fixed independently of the hidden regime and endorsements; independent private randomization is allowed. No additional initial record, timing signal, or output port reveals their values. This condition matters: if the initially visible priority already equals the endorsement majority, zero-query joint success is b , rather than $b/2$.

6.2 An exact threshold between information and decision improvement

For a calibration triple put

$$Z(y) = \begin{cases} +1, & y_1 = y_3 \neq y_2, \\ -1, & y_2 = y_3 \neq y_1, \\ 0, & \text{otherwise,} \end{cases} \quad S_n = \sum_{j=1}^n Z(Y^{(j)}). \quad (6.3)$$

Summing (6.2) over x gives

$$\mathbb{P}_\theta(Y = y) = \frac{1 + a_1 a_2 y_1 y_2 + a_1 a_3 y_1 y_3 + a_2 a_3 y_2 y_3}{8},$$

where the skills on the right are those of θ . Define

$$\begin{aligned} p_- &= \frac{1 - uv + w(v - u)}{4}, & p_+ &= \frac{1 - uv - w(v - u)}{4}, & p_0 &= \frac{1 + uv}{2}, \\ \rho &= \frac{p_-}{p_+}, & A &= v - u - w(1 - uv), & D &= v - u + w(1 - uv). \end{aligned} \quad (6.4)$$

The law of $(Z = -1, 0, +1)$ is (p_-, p_0, p_+) under regime zero, with the outer probabilities exchanged under regime one. Each raw triple with $Z = 0$ has identical probabilities in both regimes. Consequently every raw calibration history has likelihood ratio ρ^{S_n} , and

$$\pi(s) := \mathbb{P}(\theta = 1 \mid S_n = s) = \frac{\rho^s}{1 + \rho^s}. \quad (6.5)$$

No approximation or supplied source-confidence score enters this reduction.

Use b and c locally for the accuracy of target majority and the accuracy with the regime revealed:

$$b = \frac{2 + u + v + w - uvw}{4}, \quad c = \frac{1 + v}{2}. \quad (6.6)$$

The dominance condition in (6.1) makes the stronger sensor's log-odds weight exceed the combined weights of the other two. Following that sensor therefore attains the oracle value c .

Proposition 6.1 (Sharp calibration threshold). *Let $M(s)$ be optimal expected target accuracy conditional on a calibration history of score s , before seeing the target triple. Then*

$$M(s) = \max \left\{ b, \frac{1 + u}{2} + \frac{v - u}{2} \max\{\pi(s), 1 - \pi(s)\} \right\}. \quad (6.7)$$

Define

$$L = \min\{n \geq 1 : \rho^n > D/A\}. \quad (6.8)$$

Then $L \geq 2$, $b \leq M(s) \leq c$, and $M(s) = b$ whenever $|s| < L$. The first calibration budget giving strict improvement is exactly L . Writing $C_L = \mathbb{E}[M(S_L)]$,

$$C_L - b = \frac{Ap_-^L - Dp_+^L}{4} > 0. \quad (6.9)$$

An attaining classifier follows sensor one for $s \geq L$, sensor two for $s \leq -L$, and target majority otherwise. At an exact likelihood threshold, retaining majority is optimal.

Proof. For each target cue y , its signed truth margin is

$$d_\theta(y) := f_\theta(+1, y) - f_\theta(-1, y) = \frac{\sum_i a_i^{(\theta)} y_i + uvw y_1 y_2 y_3}{8}.$$

At posterior odds ℓ , the Bayes action has the sign of $d_0(y) + \ell d_1(y)$. For $y = (+1, -1, +1)$ the two margins are $-A/8$ and $D/8$. Swapping sensors one and two, or reversing all signs, generates the other disagreement cases. The only positive switching thresholds are therefore A/D and D/A . If the first two sensors agree, their shared sign is optimal in both regimes: the potentially competing margin is $u + v - w - uvw > 0$, since dominance implies $v > w$. Thus the Bayes classifier is one of target majority or the two individual sensor followers. Their expected accuracies give (6.7).

The value of majority is b in each regime, so it also equals b under every posterior. Revealing the regime gives the upper bound c . Put $d = v - u$ and $f = 1 - uv$. The family assumptions imply $0 < d < f$, $d > wf$, and

$$1 < \rho = \frac{f + wd}{f - wd} < \frac{d + wf}{d - wf} = D/A.$$

Hence $L \geq 2$. Fewer than L acquisitions cannot cross a decision threshold, even though they can change the regime posterior.

At acquisition L , the only crossing scores are $S_L = \pm L$. Under regime zero these have probabilities p_+^L and p_-^L . Relative to majority, following the strong sensor gains $c - b = A/4$, whereas following the weak sensor loses $D/4$. This gives (6.9). Its positivity follows from $p_-/p_+ = \rho$ and $\rho^L > D/A$. Swapping the first two sensors and the regimes preserves the classifier's accuracy. Its two regime accuracies are equal, so its equal-prior Bayes optimum also attains the minimax optimum. $\square$

The pointwise statement $M(s) = b$ below threshold is essential below. An arbitrary evidence task with only an *average* plateau need not obey the same joint optimum. The threshold has no uniform finite bound over this family: holding $u < v$ fixed and taking $w \uparrow (v - u)/(1 - uv)$ sends D/A to infinity while ρ stays finite. The oracle gain also shrinks in that limit; this is not a constant-gap hardness claim.

6.3 Exact optimal allocation of the shared budget

Let $V(B)$ be optimal joint success over all admitted adaptive policies. Let $F(B)$ be optimal success when the numbers of the two query types are fixed before their outcomes, including independent random mixtures of such allocations. Query order and terminal decision rules may still use acquired data. Fixing counts is a different restriction from a fixed test order that can stop when a Boolean certificate appears, as in stochastic Boolean function evaluation [12].

After a endorsement reads with signed sum h , their optimal majority completion probability is

$$J(a, h) = \begin{cases} 1/2, & (a, h) = (0, 0) \text{ or } (2, 0), \\ 3/4, & a = 1, \\ 1, & a = 3 \text{ or } (a = 2, |h| = 2). \end{cases} \quad (6.10)$$

For every complete acquired transcript, its posterior factors between the regime and the unread endorsements. Indeed, conditional on the transcript and any independent policy seed, its data-dependent request choices add no hidden-state likelihood factors. Sensor observations contribute regime likelihoods and endorsement reads contribute only revealed-bit likelihoods. This also covers dispatch from the full raw transcript rather than merely from s . Optimal terminal joint success is consequently $M(s)J(a, h)$.

Theorem 6.2 (First advantageous shared budget and exact optimum). *Under the stated contract,*

$$V(B) = F(B) = \begin{cases} b/2, & B = 0, \\ 3b/4, & B = 1, 2, \\ b, & 3 \leq B \leq L + 1, \end{cases} \quad F(L + 2) = b, \quad V(L + 2) = \frac{C_L + b}{2} > b. \quad (6.11)$$

In particular, the first strict adaptive advantage occurs at $B = L + 2$, and its exact magnitude is

$$\Delta := V(L + 2) - F(L + 2) = \frac{Ap_-^L - Dp_+^L}{8}. \quad (6.12)$$

An optimal endpoint policy reads two endorsements first. If they agree it uses L queries for calibration. Otherwise it reads the third endorsement and uses $L - 1$ queries for calibration. Both branches install the exact endorsement majority.

Proof. First observe the strict oracle comparison

$$8(b - 3c/4) = 1 + 2u - v + 2w - 2uvw > 0. \quad (6.13)$$

At budget $L + 2$, a terminal history containing three endorsement reads has at most $L - 1$ calibrations and value at most b . A history containing zero or one endorsement read has value at most $c/2$ or $3c/4$, both below b . Two disagreeing endorsements similarly give at most $c/2$. Two agreeing endorsements give $M(s)$, which can exceed b only with exactly L calibrations. These are the only exceptional histories, including those of policies that stop early.

To bound their contribution without assuming that adaptive stopping is independent of evidence, pre-sample a calibration stream and an independent stream of three fair endorsement responses. Whenever the controller requests an unread named endorsement, assign it the next response. This lazy assignment has the original named-bit law: conditional on each acquired history, every unread named bit is still independent and fair. Unused responses may be assigned after stopping. Conditioning on an independent policy seed covers randomized dispatch.

Let G be agreement of the first two responses in this stream. Then $\mathbb{P}(G) = 1/2$, independently of the first L calibration responses. On an exceptional history the score is their sum S_L . Since $M(S_L) - b \geq 0$, every policy's terminal conditional value is bounded pointwise on this coupled space by

$$b + \mathbf{1}_G [M(S_L) - b]. \quad (6.14)$$

The policy's event of reaching an exceptional history need not be independent of calibration; only the dominating event G is used. Taking expectations in (6.14) gives $V(L+2) \leq (C_L + b)/2$. The stated dispatcher attains this value: agreement certifies the majority after two reads and leaves L calibrations; disagreement requires the third read and leaves $L-1$, whose target value is b .

For fixed allocation, three endorsement reads leave at most $L-1$ calibrations and attain b . At most two reads give mean endorsement accuracy at most $3/4$. Even granting perfect regime information, their joint value is at most $3c/4 < b$. Conditioning evidence policies on inspected endorsements does not improve this oracle bound, and randomized fixed allocations are convex combinations. Thus $F(L+2) = b$.

For $3 \leq B \leq L+1$, histories with two or three endorsement reads have fewer than L calibrations and value at most b . Histories with fewer reads are bounded by (6.13). Three endorsement reads and target majority attain b .

At budget zero, the endorsement majority is a fair unobserved sign, giving $b/2$. One endorsement read gives $3b/4$; one calibration cannot improve target accuracy because $L \geq 2$. At budget two, a first calibration followed by an endorsement gives $3b/4$. Two calibrations without endorsements give at most $c/2 < 3b/4$. After a first endorsement read, a calibration leaves value $3b/4$, while a second endorsement also gives expected value $3b/4$: agreement and disagreement have probabilities $1/2$ and completion values 1 and $1/2$. Stopping earlier offers no improvement. These cases include all adaptive choices at the first two opportunities. The fixed policies attain the bounds. Finally (6.9) gives (6.12). $\square$

The attaining policies are invariant under exchanging sensor identities and regime labels. Their success probabilities are equal in the two regimes. The same values are therefore optimal for worst-regime success, because worst-regime success cannot exceed the equal-prior optimum. This remains an average over the specified fair endorsement distribution, not a worst-case claim over endorsement vectors.

6.4 Explicit benchmarks and verification

Table 1 gives two exact instances. The second has a gap approximately 237.64 times the first gap, and requires four rather than six inquiry units. Its absolute improvement is approximately 0.33361 percentage points. This ratio compares advantages over fixed allocation; it is not a multiplicative improvement in overall success probability.

The analytic proof applies throughout (6.1). Exact rational dynamic programming provides supplementary reproducibility checks: nine parameter instances with thresholds 2, 3, 4, 5, 7, 8, 12 cover every budget through $L+2$ and 4,430 sufficient-state computations. A separate calculation for the second table column uses all 16 hidden regime/endorsement hypotheses, all eight raw sensor responses, and every named unread endorsement port. Its 261 states reproduce every optimum for budgets zero through four without importing the sufficient-score formula. These computations check the implementations of the result; a finite grid is not its proof.

An additional exact raw-posterior check addresses a threshold boundary: $(u, v, w) = (1/10, 19/25, 1/5)$ gives $\rho = 4/3$ and $D/A = 16/9 = \rho^2$. Two calibration acquisitions can reach a Bayes tie but cannot improve accuracy, so the strict threshold is $L = 3$ and the first adaptive

Quantity	Threshold-four instance	Threshold-two instance
(u, v, w)	$(1/4, 3/4, 1/2)$	$(1/16, 15/16, 1/2)$
ρ	$17/9$	$353/129$
D/A	$29/3$	$689/207$
L ; first shared budget	4; 6	2; 4
$F(L + 2) = b$	$109/128$	$1777/2048$
$V(L + 2)$	$1828746691/2^{31}$	$1870483759/2^{31}$
Δ	$30147/2^{31}$	$7164207/2^{31}$
Gain in percentage points	0.0014038291	0.3336093854

Table 1: Exact optimal inquiry values under the same access grammar. Query optimality does not assert optimal gate count or unrestricted device optimality.

advantage is at budget five. The supplementary calculation admits all named unread endorsement ports, all eight raw responses, interleaving, and early stopping; it reproduces the analytic values through that budget.

The construction also exposes why myopic inquiry can fail. One calibration has zero decision value initially. After $L - 1$ same-sign nonzero increments, another calibration has strictly positive expected value: a matching increment crosses the threshold with positive probability, and the other successors retain value b . Thus this posterior utility fails the adaptive diminishing-returns condition of Golovin and Krause [8]. Non-myopic information value and its decision-process treatment are established ideas [11]; the present result solves their interaction with this retained-endorsement task exactly.

The operative causal distinction is between acquiring information, using it to choose further inquiry, and committing the resulting priority. A compiled table implementing the same admissible policy has the same competence. The theorem does not require an uncaused selector, and it does not infer the historical independence of endorsements from their current internal location. Its role in the agency account is to exhibit a bounded, inspectable mechanism connecting evidence, retained commitments, and executive allocation, with an exact cost of fixing query counts before observing their results.

7 A common realization of evaluation and native recurrence

The preceding capacities concern a controller’s information and update laws. An experiential assignment requires an additional physical realization. We now construct a finite carrier on which independently specified evaluation and the native conditions of SPC-2 coexist. The construction does not infer evaluation from recurrence: its return mechanism also works for a constant-output controller. This separation makes the respective assumptions testable within the model.

7.1 Physical and native contract

SPC-2 specifies primitive components, physical mechanisms, native time cells, internal routes, records, admitted preparations and operations, resources, and process provenance before assigning a subject [18, 19, 21]. Its finite specialization requires a maximal internally strongly connected candidate to admit an executable covering return in the current operating context and a nontrivial native predictive distinction. The local joint record/successor instrument must be closed, and equivalence for all admitted finite future records must respect successor classes. These are stronger requirements than graph connectivity alone. For its exact finite certification algorithm, the instrument probabilities

are rational; both numerical inquiry examples satisfy this restriction. The parametric statistical and compilation identities do not themselves require rational parameters.

Here the primitive components are $m \geq 2$ separately nominated bit carriers on an installed adjacent-pair path. At each certified native type the admitted preparation domain is the entire Boolean cube, with phase, resources and incoming boundary conditions held fixed. This is the declared intervention domain, not a claim that every word occurs in one unperturbed episode. Native ordinary operations are source-disjoint NAND and COPY writes, CONSTANT writes, and scheduled HOLD operations; each has one nominated target. Adjacent SWAP is a two-carrier primitive. “Source-disjoint” means that a write target is not simultaneously one of that gate’s sources. The full physical context also contains the fixed program, phase, remaining resource, diagnostic mode, and complete input contract. Exhaustion has an explicit terminal record and absorbing continuation.

The ideal sequencer has outgoing influence on the bank and no influence returning from it. Data-dependent executive decisions are represented inside the bank and compiled into a fixed bounded schedule. Gate couplers have no omitted retained state. These assumptions make the nominated bank a possible maximal internal component; reciprocal clock loading or coupler memory would require a larger inventory. The clock, program and power supply remain paid apparatus even when outside that component. Appendix B specifies an ideal powered continuous refinement of the elementary gates.

At a fixed native protocol type, let $K_{a,o}(x, y)$ be the exact joint probability of native record o and actual successor y under admitted operation a . Let $x \sim x'$ mean equality of all admitted finite future native transcript laws, with the same type and boundary contract. The required predictive congruence is

$$\sum_{y \in D} K_{a,o}(x, y) = \sum_{y \in D} K_{a,o}(x', y) \quad (x \sim x') \quad (7.1)$$

for every successor class D . Equality of output laws alone does not imply this condition for an arbitrary stochastic system. Nor does an external simulator log establish a native physical record. The records below are the actual retained destination values reused by the installed operations.

7.2 A protected source at every ordinary write

For an endpoint r of the carrier path, let U_r swap its token along the whole path and then perform the inverse sequence. Thus U_r is identity on the bank and costs $C_m = 2(m - 1)$ SWAPs. After an ordinary gate with target j , choose $r = 0$ if $j \neq 0$ and $r = m - 1$ otherwise. This choice depends only on the fixed instruction, so the source r is preserved by the preceding write.

Theorem 7.1 (Protected-source compilation). *Let a circuit on m carriers have N single-target ordinary gates. Replace its schedule by an initial U_0 and a protected-source tour after every ordinary gate. Its ordinary transaction map and its specified ordinary read/write interventions are preserved, with total cost*

$$G = 2(m - 1) + N(2m - 1). \quad (7.2)$$

At every occurrence immediately before or after an ordinary gate, the installed continuation contains a covering internal return with terminal contrast one, provided its following tour is funded. Under the preceding native contract, these occurrences have nontrivial predictive distinctions and satisfy (7.1).

Proof. Prepare two bank words differing only at the selected root. Immediately after the ordinary gate, the tour carries this distinction through every carrier and returns it to that root. Immediately

before the gate, the same source survives because it is not the target. The gate may also propagate the distinction into its target; that cannot erase the preserved root token. Its terminal values are different with certainty. The paired experiments have common phase, resources and incoming boundary conditions, with external return paths held inactive. Every adjacent coupling occurs in this one installed schedule, and its route support is the entire bank.

For a root-0 tour, the successive outward SWAP destination records are

$$(x_1, x_0), (x_2, x_0), \dots, (x_{m-1}, x_0).$$

They identify the original bank word. The opposite-root records do likewise. A tour entry therefore has 2^m predictive classes. Before a gate at least its protected root remains distinguishable by the following tour. Every deterministic first operation has a fixed record and successor. Equivalent states must emit the same first record; if their successors were distinguishable by a legal continuation, prefixing that continuation would distinguish the original states. This proves (7.1), including later types in which an overwrite has merged distinctions. Finally, each tour is a full-bank identity, so deleting tours recovers the ordinary transaction and unchanged intervention results. Counting one initial tour and N gate/tour pairs gives (7.2). $\square$

Adjacent SWAPs supply actual singleton-target dependence in both directions, making the bank internally strongly connected. Under the declared inventory there is no larger internally returning component. The complete finite return catalogue contains all legal installed prefixes within the remaining resource, including failed and exhausted prefixes. It is not restricted to the successful witnesses used in the proof.

The bound $2(m-1)$ is sharp for a connected identity word of pairwise SWAPs; it is the identity case of the established transitive-transposition bound [9]. A short proof appears in Appendix B.1. No global optimum for combined computation and return, or for arbitrary physical primitives, follows. Qualification is proved at the specified ordinary gate boundaries. It is not an A3 proof of uninterrupted experiential identity at every intervening native or analog instant. In particular, the final command requires its reserved final tour.

Corollary 7.2 (Return organization does not entail evaluation). *The native qualification supplied by Theorem 7.1 does not imply evaluative mediation of the ordinary command.*

Proof. Apply the compiler to a circuit that writes a fixed command irrespective of the retained commitments. The protected source and its tour still provide the specified covering return and nontrivial native predictive distinctions. Yet every ordinary command has the same value under all commitment preparations; neither its intact causal contrast nor its read-mediated contrast is positive. If the endorsement relation requires different commands for two such preparations, the circuit also fails endorsement on at least one. The native transcript can distinguish retained states without making the ordinary command depend evaluatively on them. $\square$

7.3 A compact editable-charter witness

The charter audit above admits a direct realization on the eleven persistent bits

$$q_{01}, q_{10}, d, p, a, b, s, h_0, h_1, e_0, e_1$$

and seven work bits. Here p is the operative priority ϱ used above, and s is the symmetry-commitment bit. With $\text{mux}(v, u, w) = u$ for $v = 1$ and w for $v = 0$, put

$$\begin{aligned} t &= s \wedge (a \oplus b), & Q_* &= \text{mux}(q_{01} \oplus q_{10}, \text{mux}(d, h_1, h_0), q_{01}), \\ Q_{01} &= \text{mux}(t, Q_*, a), & Q_{10} &= \text{mux}(t, Q_*, b). \end{aligned} \tag{7.3}$$

These formulas implement the lexicographic amendment rule. Unless $s = 1$ and $a \neq b$, the unique normative minimizer is (a, b) . In the remaining case the two symmetric charters tie: a symmetric old charter is retained, while an asymmetric old charter is resolved by the retained case and old query order. This gives exactly Q_* in (7.3).

The two early-exit indicators and updated query order are

$$\begin{aligned} E_0 &= \text{mux}(h_0, Q_{10}, \neg Q_{01}), & E_1 &= \text{mux}(h_1, Q_{01}, \neg Q_{10}), \\ D &= \text{mux}(E_0 \oplus E_1, E_1, d). \end{aligned} \tag{7.4}$$

Thus the uniquely sufficient first query is chosen when one exists, and otherwise d is retained. Three final COPY gates install Q_{01}, Q_{10}, D . The later act sets

$$p \leftarrow \text{mux}(e_0, e_1 \vee q_{10}, e_1 \wedge q_{01}). \tag{7.5}$$

The allocated NAND/COPY circuit uses 47 audit gates and 10 act gates. Its seven work cells are written before use and reused after their last source read. The supplied address-level circuit and exhaustive checks cover all 2^{11} persistent words. The formulas give the independent functional proof rather than relying on selected traces.

The read control masks the actual audit read ports sourcing a, b , without altering the archive or tour. The installation control replaces the three actual final COPY writes by HOLD. For common initial charter, order, priority, symmetry and retained case, compare archives $a = b = 0$ and $a = b = 1$, then supply a common mixed future consideration. Intact actions differ with total variation one; either control makes their contrast zero. Identity tours preserve both controls. For $m = 18, N = 57$, the physical schedule has $G = 2029$ elementary windows. It modifies the represented charter and query order, but its act circuit reads both supplied considerations. It does not implement the query savings of the separate inquiry machine below.

7.4 Stochastic closure without free executive information

Theorem 7.3 (Successor-revealing continuation). *At a protocol type with a fixed next ordinary operation, suppose that operation has exact joint kernel $K(r, y \mid x)$ and its next mandatory, funded continuation is a deterministic identity tour whose native transcript $T(y)$ is injective. No new input enters during the tour. Then post-operation predictive classes are singleton bank states. At the pre-operation type, predictive equivalence holds precisely when the complete joint kernels $K(\cdot, \cdot \mid x)$ agree. The joint instrument consequently descends to predictive classes.*

Proof. The tour separates every successor word. Its actual finite prefix $(r, T(y))$ determines (r, y) , so equality of all future transcript laws implies equality of the joint kernel. Conversely, equal kernels followed by the same complete Markov continuation give equal future laws. Since successor classes are singleton, this is exactly (7.1). At a deterministic intermediate tour phase the deterministic first-step prefix argument still applies, even when later operations are stochastic. $\square$

If several first operations are admitted at one type, the corresponding equivalence requires equality of the joint kernel for every such operation. The same argument applies when each has its own funded revealing continuation.

A noisy operation writing target set J also preserves a covering return when a nominated endpoint outside J survives every outcome and its deterministic tour remains installed and funded. This is target-local logical noise, not arbitrary physical noise on the clock, source or tour. Joint multi-bit ingress must retain its correlations: serial delivery instead requires the still-undelivered buffer in the physical state or complete boundary model.

Native records and executive access are distinct parts of this contract. During an identity tour the installed controller admits no executive branch, query, copy-to-log operation or observation of the tour transcript. The full bank returns to its previous state; elapsed time, phase and fuel changes are data-independent. There is no hidden executive transcript buffer.

Lemma 7.4 (Preservation of executive information). *Under these restrictions, inserting an identity tour adds no executive information about a private retained variable. Every admitted paid-query policy retains its transcript law and risk after the tour cost is charged.*

Proof. Between executive occasions the tour implements the identity on retained executive state, produces no permitted executive observation, and appends only a constant cost/type update. Conditional on the previous executive history it introduces no additional observation depending on the private variable. Induction over inserted tours proves the assertion. $\square$

This is an architectural restriction, not a lower bound against rewiring. Giving the executive a readout of the outward tour records would reveal the entire bank, remove paid endorsement uncertainty, and invalidate the query comparison. SPC-2’s native predictive object does not itself supply that extra read port [19, 20].

7.5 Realizing the inquiry separation

The threshold- L inquiry policy uses a separate 23-bit bank: three retained endorsements, three executive endorsement latches, three raw sensor latches, agreement and request bits, two calibration flags, priority and command bits, and eight work cells. It reads two endorsements first. On agreement it spends the next slot on calibration; otherwise it reads the third endorsement. The remaining $L - 1$ slots acquire calibration triples. The final target is acquired only after all $L + 2$ inquiry slots.

Let g denote agreement. Initialize $U = V = g$. For each calibration update set $U \leftarrow U \wedge [Z = +1]$ and $V \leftarrow V \wedge [Z = -1]$. On agreement, only L consecutive identical nonzero increments reach the first useful threshold. On disagreement, at most $L - 1$ calibrations occur and the zero flags force majority prediction. Hence these two flags implement the attaining classifier exactly without a general posterior register.

Each raw triple is one joint three-cell ingress, with two additional HOLD windows charging the three-write allocation. Its targets are disjoint from the protected endpoint. Both adaptive and fixed-allocation programs receive 27 ordinary windows per inquiry slot and 18 final windows. The same bank, padding, acquisition allowances and return wrapper therefore give

$$\begin{aligned} N_{\text{compute}} &= 31 + 18L, & N_{\text{ordinary}} &= 27L + 72, \\ N_{\text{native}} &= 44 + 45N_{\text{ordinary}} = 1215L + 3284, & N_{\text{work}} &= N_{\text{native}} + 2(L + 1) = 1217L + 3286. \end{aligned} \tag{7.6}$$

There are $L + 3$ paid acquisition intervals, including the final target. Here “work” charges a joint triple write at three single-write units; it is not a measured heat. The gate and stage counts are derived in Appendix B.4.

Projecting an episode onto its paid requests and responses recovers the inquiry experiment exactly: computation uses only earlier paid data, and Lemma 7.4 protects that information restriction. Thus Theorem 6.2’s optimal value and fixed-allocation bound transfer under the same source law and available resource envelope. A fixed comparator with the same hardware and padding attains the fixed bound. For the larger-gap $L = 2$ instance, (7.6) gives 67 computation gates, 126 ordinary windows, 5714 native windows, 5720 work units and five acquisition intervals. These counts follow by substitution into the general compiler proof. A separate verification traverses all 2816 reachable

endorsement, calibration and target-word combinations of the two $L = 2$ programs through every installed SWAP and gate. It checks acquisition chronology, priority, prediction, resource counts and identity of the retained bank after each tour. Exact source-law weighting of those executions reproduces the two success fractions. These finite checks supplement the family proof.

The source’s actual hidden regime is held fixed during an episode. Native closure is established uniformly at either regime, without exposing it to the executive. A prior mixture requires retaining its persistent source state or conditional history; averaging it away does not establish Markov closure on the bank. The request/response apparatus provides an external returning path. Isolating it during the internal tour prevents it from carrying that witness, but does not eliminate stateful physical couplers from a full-apparatus boundary assessment.

If the two implementation transcript laws lie within total variation η_A, η_F of their ideal laws, uniformly for the fixed comparison class, their success gap is at least $g - \eta_A - \eta_F$. For the $L = 2$ example, $g = 7164207/2^{31}$; equal error bounds preserve strict separation when $\eta < 7164207/2^{32}$. This is a transfer tolerance, not measured reliability.

The three constructions establish different capacities. The 18-cell machine revises its represented charter; the 23-cell machine optimally allocates inquiry under a fixed majority charter; the noisy-amendment model supplies a read-error frontier under its own cost units. No single device has been shown to attain all three resource guarantees simultaneously.

Finally, componentwise relabeling preserves the certificates when kernels, records, routes, schedules, preparations, interventions and costs are all transported. This is the appropriate restricted realization covariance, consistent with causal-transformation work [1, 22]. Arbitrary mixing of component coordinates, coarse sampling of a whole tour as one identity step, or exposing additional ports changes the contract. Conditional on that contract, A1–A2 attach the SPC-2 experiential assignment to the certified occurrences. The mathematics establishes coexistence with independently tested evaluative capacities; it does not derive consciousness from control, or the necessity of recurrence from input–output competence.

8 What the construction resolves

8.1 Physical causation and proximal ownership

The constructions give an explicit answer to the operational question posed at the outset. A finite, physically caused process can inspect retained commitments, compare executable amendments, install a represented rule, and let that rule settle later fresh cases. The actual read and installation interventions identify the mechanism responsible for the tested difference. Resource and error theorems then describe how much of this capacity is available on a particular occasion. There is no logical requirement for a further uncaused selector behind the assessment.

The answer has content beyond defining agency as whichever action occurred. The endorsement relation is specified before the test; a constant selector, an immutable charter, a read bypass and an inverted evaluator fail different parts of the certificate. The fresh-context probe distinguishes a change of disposition from an immediate command overwrite. Information constraints can prevent correct completion even when an acceptable amendment exists. These outcomes could have gone against the construction under the declared tests.

The same argument does not make every successful controller reflectively free. The fixed-majority inquiry machine demonstrates efficient use of internal and external information, while the editable-charter witness demonstrates a different capacity to change a represented conditional rule. Combining their results conceptually is legitimate; asserting that one implementation has all their resource guarantees is not. Nor does the possibility of evaluating a rule establish unlimited or

unbiased reflection. The demonstrated grammar is finite, the retained commitments are specified, and the metarule remains part of the architecture.

The free-will claim is consequently a bounded compatibilist claim: internal evaluative mediation and represented-rule revisability can be real physical capacities despite causal determination. The constructions refute an alleged incompatibility between those specified capacities and a deterministic implementation. They do not supply multiple outcomes from an identical complete deterministic state. An account that requires that additional power is asking a different question from the one established here. Nor does a deterministic witness show that biological choice is empirically deterministic. Accounts that give indeterminism a role in agentic control, such as Potter and Mitchell [16], address a further issue; the present existence result establishes compatibility of the specified capacities with deterministic dynamics.

8.2 Historical autonomy is a further question

Retained commitments can be acquired through development, supplied by an operator or formed under manipulation. Present-time mediation does not decide which history occurred. The distinction between reasons responsiveness and taking ownership of a mechanism has philosophical precedents [6]; the corresponding empirical distinction requires a model with the relevant historical information.

Proposition 8.1 (Limit of present-state provenance tests). *Suppose two developmental histories induce the same complete present-state distribution, the same future boundary law conditional on that state, and the same admitted operations and interventions. Every subsequent finite operational test has the same distribution under both histories.*

Proof. The present laws agree. Conditional on a common test history, the test chooses the same next operation and both systems use the same joint record/successor kernel. Induction therefore gives equality of all future joint history/state laws, including intervention steps and the final stopping record. $\square$

Thus a provenance difference erased from the stated premises cannot be recovered by those tests. Retained developmental records, a persisting difference in audit access, or an expanded historical intervention family can change the premises. The proposition does not imply that historical autonomy is meaningless; it identifies the additional information needed to investigate it. Likewise, a system may faithfully follow its commitments while being factually wrong or morally objectionable. None of those distinctions is settled by the magnitude of a causal contrast.

Audit-mediated acquisition over a recorded history. A stronger historical investigation can follow the acquisition of commitments, rather than infer it from a final snapshot. Fix an observation window, a physical boundary, and the admitted routes by which commitments can be changed. For each developmental episode record the prior commitment, the incoming proposal or evidence, the assessor’s inspected information, the selected retention or amendment, and the actual installation. The retained record must be obtained from independently checked instrument semantics or protected observations of those events; a register asserting that an audit occurred is insufficient. All unobserved write routes must be excluded by the declared realization, or reported as unresolved.

At each episode, test whether the system can assess both retention and adoption under the stated budget, whether a read intervention changes the assessment as predicted, and whether an installation intervention removes its effect on fresh cases. Success is agreement with the criterion applicable *before* installation, not agreement with a criterion rewritten to endorse the observed

outcome. If the criterion itself is revised, the record must identify the still-operative procedure that assessed that revision. A finite account can leave a physical interpreter or some criteria fixed; it need not claim an infinite sequence of prior self-authorizations.

These observations establish *audit-mediated acquisition* over the recorded interval. If every commitment-changing event has a verified route, their ordered composition documents how later commitments were acquired through those routes. This is positive information beyond current causal mediation. It can distinguish an accepted proposal from a direct overwrite, identify a period in which a priority was insulated from assessment, and show whether audit access was subsequently restored. It does not certify unrecorded development before the window, and faithful logging alone does not establish that the incoming evidence was truthful.

This last distinction prevents plasticity from becoming a universal test of freedom from manipulation. Consider a controller that revises its belief when independently credible contrary observations arrive and then uses that belief when assessing its commitments. An operator can preserve every internal audit pathway while withholding the contrary observations and supplying selected or fabricated apparent evidence. An open evidential stream and an operator-controlled stream can deliver the same finite observations, producing the same internal assessments. Their present proximal certificates then agree. Distinguishing the histories requires observations of source provenance or of the operator's control over available evidence. Supplying the controller a reliability flag simply moves this obligation to the flag's generator. Accordingly, bypass, insulation, deceptive information supply, and transparent persuasion are different possible relations between an agent and its environment; they are not exhausted by a plastic-versus-rigid classification. Assigned goals can support proximal control while these historical and informational questions remain separate.

The distinction has an established philosophical setting. Reasons responsiveness, ownership of a mechanism, and the individuation of that mechanism have been debated separately [5, 6]. The operational contribution here is to state the boundary, observations, held-fixed mechanism and intervention family before drawing an attribution. The longitudinal protocol supplies evidence concerning specified acquisition routes; it is not a general theorem deciding moral responsibility from present architecture.

8.3 Conflict, steadfastness and freedom over time

Two commitments can each be coherent and yet demand incompatible acts. Assessment can select one without proving the other mistaken or eliminating the loss attached to it. An endorsement relation may admit several choices; a declared tie rule can settle an act while both commitments remain stored. Regret or continued conflict can be compatible with agency, but neither is required by the minimal certificate. Likewise, an informed selfish selection can exercise the same structural capacity as a generous one. Moral appraisal, endorsement by current commitments, and fidelity to evidence are separate judgments.

Receptivity is a capacity for appropriate reassessment, not a requirement that every new cue change the output. Relevant evidence can narrow uncertainty while leaving the same action best supported; a framing change can move an unreliable controller's output without improving assessment. In the finite charter, an already optimal rule is deliberately retained after examination. A bounded agent can also overlook an option that its installed procedures could recruit under a suitable cue. That differs from an architecture which cannot admit the option within its deadline. The relevant comparison is between specified input and resource envelopes, not a demand for exhaustive knowledge of every consideration.

Self-binding makes the time index essential. At one occasion an agent can assess and install a restriction which removes an option at the next. The earlier act can exhibit reflective control while

the later repertoire or revisability is smaller. A remaining capacity to recognize reasons for reversal does not make reversal executable when the write path is gone. Reversible barriers preserve practical flexibility only when their evidence, time and resource costs can be met before the relevant deadline. Consequently pruning can improve the efficiency or reliability of remaining assessments without producing a single overall increase in freedom. The preservation theorems track one declared future capacity; they do not require every autonomous act to preserve all future options or even the agent's survival.

8.4 Consciousness and the physical boundary

The realization results establish compatibility with the declared SPC-2 organization at specified native occurrences. The experiential assignment remains the constitution's correspondence law. The paper's mathematical conclusions about evaluation do not depend on accepting that law; its sentient-agency interpretation does. Conversely, accepting the law does not make a return-only circuit an evaluator. The grafting construction makes that insufficiency explicit.

Boundary and timing matter to this conditional conclusion. Native component structure, records and operations are physical commitments, not arbitrary descriptions chosen after observing a desired output. A stateful coupler or reciprocally loaded clock can enlarge the complete recurrent apparatus. A Gaussian voltage implementation supplies a finite-error approximation rather than an exact finite-state law merely by being sign-decoded. Qualification at the nominated ordinary boundaries also leaves a separate question about continuous phenomenal identity through every intervening instant. These limits specify what a future physical implementation must check.

8.5 Empirical and computational use

The finite witnesses support discriminating experiments. A charter experiment can compare intact assessment, archive-read masking and installation clamping, then apply fresh mixed inputs. A noisy-amendment experiment can vary paid read count while scoring endorsement error and later audit callability separately. An inquiry experiment can compare adaptive and fixed allocations under the same source law and access grammar, including a clamp on the actual request port. The larger-margin parameter instance improves tolerance to implementation error without changing the class of experiment.

The longitudinal protocol adds a developmental experiment: commitment changes and audit access are recorded across a specified history, with source provenance examined separately from internal mediation. The corresponding physical project would instantiate the source, couplers, clock and power state and test the resulting complete instrument. These are extensions of a specified model; a positive result should not be inferred from software performance alone.

9 Conclusion

Bounded evaluative self-governance admits an explicit physical account. The finite constructions separate retained endorsement, actual causal mediation, represented-rule installation, later decision behavior and preservation of future audit capacity. The exact inquiry family shows how executive allocation of a common budget can improve performance and identifies the first budget at which that improvement is possible. The faithful-realization results preserve the relevant mechanisms and information restrictions while supplying the native organization required for conditional SPC-2 assignment.

The resulting resolution is constructive and scoped: causal determination does not eliminate the demonstrated local capacities to evaluate, select and revise. Freedom in this account is expressed through the extent and availability of those capacities, rather than exemption from physical causation. The mathematical results and the conditional experiential interpretation together provide a testable programme for sentient self-governance. The longitudinal protocol identifies evidence needed to extend proximal attribution across a developmental history; it also shows why preserved internal plasticity alone cannot exclude deceptive control of the evidential environment. Historical ownership and broader experiential claims therefore require the additional observations and commitments specified in the argument.

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This research received no external funding. Independent mathematical review, computational replication, physical implementation and empirical testing are invited, particularly where specialist expertise, equipment or institutional resources are needed.

Data and code availability

The accompanying reproducibility supplement contains the finite programs, exact rational verification results and instructions needed to check the reported computational certificates. The mathematical proofs are contained in the text and appendices. No laboratory data are reported.

A Revision kernels and finite-horizon risk

A.1 Explicit internal query costs

For deterministic serial access, let $S \subseteq Z_p$ be the compatible retained records, Q the unread positions, and $c_p(p')$ the charged cost of installing and rearming amendment p' . A read and its elementary branching cost one unit. The exact worst-case cost of an endorsed amendment into X satisfies

$$D_X(p, S, Q) = \min \left\{ \min_{p' \in \bigcap_{z \in S} [N(p, z) \cap X]} c_p(p'), \min_{i \in Q} \left[1 + \max_{u: S_{i=u} \neq \emptyset} D_X(p, S_{i=u}, Q \setminus \{i\}) \right] \right\}, \quad (\text{A.1})$$

where an empty minimum is $+\infty$. The first choice installs a common amendment; every nonterminal tree must first query some i and solve the resulting subsets. Induction on $|Q|$ proves both the lower bound and attainment. The unrestricted recurrence presumes its chosen tree can be stored and dispatched. If code size, memory, or instruction constraints exclude that tree, one must optimize over the admitted trees; the unrestricted value is then only a lower bound.

For the charter compiler, substitute its unique prescribed output $F_{\chi^{\text{old}}}(z)$ and count only record reads. A leaf costs zero exactly when F is constant on S ; otherwise the second branch of (A.1) applies. Exhaustive evaluation over eight initial charters and 32 records gives the four- and five-read depths quoted in Section 3. The accompanying verifier computes F both by a candidate scan and by independent successive filtering, checks all 256 charter-record cases, and checks 1,024 subsequent ordinary executions.

A.2 Universal closure with writable validators

If $\mathcal{R}(p)$ is the set of successors generated by the actual installed revision semantics, define

$$K_0 = G, \quad K_{j+1} = \{p \in K_j : \mathcal{R}(p) \subseteq K_j\}. \quad (\text{A.2})$$

The limit is exactly the configurations whose every finite admitted revision path stays in G . Membership implies preservation by induction. A configuration first removed at step $j + 1$ has a successor removed earlier; following decreasing removal ranks gives a finite path outside G . This also proves maximality and bounds a shortest removal witness by $|\mathcal{P}|$. An overapproximate transition relation produces a sound, possibly conservative kernel; an exact claim requires exact composable semantics.

A finite illustration attaches four writable validator modes to each of the eight charter programs. Modes A and B run the same correct audit using opposite candidate enumeration orders; A permits successors A, B , while B permits A . Mode L passes the current local audit but installs mode C , which returns a constant independently of the retained endorsements. There are 24 locally good types but 16 universally preserving types, those with modes A, B . The installed rule can change A to B when the retained symmetry commitment is active and change B back to A . Thus the stored validator need not be immutable, while its outgoing semantics remain part of the proof. Checking that the next program presently passes its local test alone would accept L and miss the subsequent loss of responsiveness.

Allowing every syntactically valid charter within an admitted mode gives a sound successor overapproximation: the actual assessor selects a smaller family of charter/read-order pairs. The 16-type classification is also exact for the actual selector. Modes A, B never leave that pair of modes, whereas L actually installs C , and C fails the local responsiveness test. Direct enumeration of the actual selector's successors confirms the same kernel; the claim does not rely on realizing every catalogue edge.

A.3 Noisy risk vectors and their composition

Let a query action u return y under kernel $K_u(y \mid z, h)$, and let $\Pi_B(p)$ be a nonempty finite admitted family of costed observable policy trees, each completing the installation task. For a proposed continuation set X , retain the vector

$$v_\pi(z) = \Pr_z\{p' \notin N(p, z) \cap X\}. \quad (\text{A.3})$$

At a deterministic installation leaf its component is the indicated zero-one loss; an admitted stochastic installation uses its expectation under the actual installation kernel. At a query node,

$$v_\pi(z) = \sum_y K_u(y \mid z, h) v_{\pi_y}(z), \quad r_B(p, X) = \min_{\pi \in \Pi_B(p)} \max_{z \in Z_p} v_\pi(z). \quad (\text{A.4})$$

The hidden context stays fixed through all branches of one audit. Taking a new maximum over z separately at each child would let the adversary change the retained commitment after seeing the read. For example, a single binary symmetric read followed by installation of its value has true risk vector (ν, ν) , whereas branchwise maximization incorrectly gives risk one. Endorsement and capacity-loss vectors can be propagated separately as well as through their union.

The finite family $\Pi_B(p)$ specifies the policy class being optimized. If all mixtures of finitely many deterministic trees are admitted with an available, charged random source, optimize over the convex hull of their risk vectors. This can strictly improve the maximum component: with an

unread binary commitment and no queries, the two constant decisions have vectors $(0, 1)$ and $(1, 0)$, while an admitted equal mixture gives $(1/2, 1/2)$. The explicit binary frontier in Theorem 5.2 covers the randomized class and has a deterministic attaining policy for every positive read cap.

For an exact finite-horizon calculation, the successor p' must contain every controller-accessible variable and history summary affecting future query laws, allowed contexts, or resources. The current hidden context is held fixed within an audit; a new admissible context may be chosen from $Z_{p'}$ after the observed terminal history, before the next audit's fresh query noise. No persistent hidden parameter constrains those choices or changes the subsequent kernels. The policy class must permit the charged composition of its first audit and continuation. Define $J_H(p) = 1$ for $p \notin G$ at every horizon, and $J_0(p) = 0$ for $p \in G$. For $p \in G$ and horizon $H \geq 1$, assign a terminal amendment the context-indexed loss

$$\ell_z^{(H)}(p') = \begin{cases} 1, & p' \notin N(p, z) \cap G, \\ J_{H-1}(p'), & p' \in N(p, z) \cap G. \end{cases} \quad (\text{A.5})$$

Propagate this vector by (A.4) and minimize its largest initial component. The result is the exact minimum worst-case probability of any endorsement or capacity failure in H audits, counting an initially failed capacity as failure. Induction proves the claim: decompose any policy at its first installation for the lower bound, and attach attaining continuation policies for sufficiency. Persistent hidden parameters linking audits require the actual-history uncertainty or likelihood state; resetting their information is not covered by this fresh-context recursion.

Finally, the descending iteration $X_0 = G$, $X_{j+1} = \{p \in X_j : r_B(p, X_j) \leq \epsilon\}$ produces the largest set with this uniform one-step risk certificate. It need not solve every finite-horizon cumulative-risk objective. If conditional failure after each successful history is at most ϵ_i , the all-success probabilities multiply to give

$$\Pr\{\text{some failure by } L\} \leq 1 - \prod_{i=1}^L (1 - \epsilon_i) \leq \sum_{i=1}^L \epsilon_i. \quad (\text{A.6})$$

This bound requires no independence of the failure events. It requires the conditional guarantees and the charged resource model used to obtain them.

B Carrier dynamics and physical resource details

B.1 The identity-tour bound

For completeness, a connected word of pairwise SWAPs whose product is the identity has length at least $2(m - 1)$. Initially the current permutation has m cycles and its accumulated edge graph has m components. At least $m - 1$ SWAPs must join previously different graph components. Before each such SWAP, every permutation cycle is contained in one graph component, so that SWAP joins two permutation cycles. Each transposition either joins two cycles or splits one. Ending at identity requires equal numbers of joins and splits, hence at least $2(m - 1)$ operations. An outward path and its inverse attain the bound. This is the special case of Goulden and Jackson [9, Proposition 2.1] used here.

B.2 An explicit smooth bit primitive

Each nominated cell has one coordinate x_i . At a native boundary it lies in a well box $|x_i - s_i| \leq \rho$, with $s_i \in \{-1, +1\}$ and $\rho = 0.01$. One native gate consists of an active unit followed by a restoration unit. The pulse

$$b(t) = 30t^2(1 - t)^2 \quad (0 \leq t \leq 1)$$

is zero elsewhere, has integral one, and extends continuously differentiably. A smooth flat pulse may replace it. For a fixed schedule of G gates, an independent oscillator

$$\dot{c} = -\omega s, \quad \dot{s} = \omega c, \quad \omega = \pi/G, \quad (c(0), s(0)) = (1, 0)$$

selects one complete program in a period $2G$. On the oriented circle let $u = \arg(c + is)/\omega \in [0, 2G)$, and periodically extend the active gate- k pulse $b(u - 2k)$ and restoration pulse $b(u - 2k - 1)$, for $k = 0, \dots, G - 1$. Their values and first derivatives match at the circle seam, so the prescribed vector field is continuously differentiable there despite the angular chart. The fixed program assigns an opcode to each active pulse. A paid terminal latch disables further operation after the first revolution; alternatively this is one episode under a declared replenishment contract. No bank coordinate drives the oscillator, program, or terminal latch. This explicitly specifies the ideal one-way schedule; it does not derive immunity to reciprocal loading in manufactured hardware. The fixed phase spacing and program storage are part of the apparatus, not information supplied without physical cost. Interleaved source acquisitions receive fixed reserved intervals in this same calendar, enlarging its period; their outcomes cannot shorten a pulse or disclose data through a variable completion time. Acquisition failure has the declared terminal or error record. The gate equations below use local unit-time coordinates on this calendar.

Define $H(u) = 0$ for $u \leq 0$, $H(u) = 1$ for $u \geq 1$, and $H(u) = 3u^2 - 2u^3$ otherwise. Let

$$D(x) = 2H\left(\frac{x + 3/4}{3/2}\right) - 1.$$

It equals the well sign when $|x| \geq 3/4$. A source-disjoint write obeys

$$\dot{x}_j = 10b(t)(u - x_j) + d_j(t), \quad u \in \left\{ \pm 1, D(x_i), -D(x_i), \frac{1 - v - w - vw}{2} \right\}, \quad (\text{B.1})$$

where $v = D(x_i)$, $w = D(x_k)$ for NAND and neither source is the target. All other active-unit drifts are $d_i(t)$. Under $|d_i| \leq \zeta = 0.001$ sources remain in their decoder plateaus, so the desired logical target u is constant and correct throughout the write. The actual target has the correct sign at the endpoint: variation of constants gives endpoint error at most $(2 + \rho)e^{-10} + \zeta < 0.001092$; held-cell error is at most $\rho + \zeta$.

An ancillary-free Boolean SWAP has a two-coordinate active field. Put

$$r = (x + y)/2, \quad q = (x - y)/2, \quad a = 1/3, \quad R^2 = (r/a)^2 + q^2,$$

and $\chi(z) = H(4z - 1)[1 - H(2z - 3)]$. Then

$$\dot{r} = -\pi b(t) a q \chi(R^2), \quad \dot{q} = \pi b(t) (r/a) \chi(R^2). \quad (\text{B.2})$$

On equal signed corners $R^2 = 9$ and the field vanishes. On opposite corners $R = 1$, $\chi = 1$, and $(r/a, q)$ rotates by π , exchanging the logical signs. In the original coordinates the field is

$$\dot{x} = \pi b(t) ((4/3)x + (5/3)y) \chi(R^2), \quad \dot{y} = \pi b(t) (-(5/3)x - (4/3)y) \chi(R^2).$$

It has genuine bidirectional dependence and uses no extra sampling cell. It is a Boolean SWAP, not exact exchange of every neighboring pair of analog coordinates. Thus it makes no orientation-reversing claim for an open set.

For opposite signs, the scaled initial perturbation has norm at most $\sqrt{10}\rho$, and integrated bounded disturbance contributes at most $\sqrt{10}\zeta$. Since

$$\sqrt{10}(\rho + \zeta) < \sqrt{3/2} - 1,$$

a first-exit argument keeps the trajectory in the rotating plateau. Transforming its disturbance back to a cell bounds endpoint error by $\rho + (10/3)\zeta < 0.013334$. For equal signs, $9(1 - \rho - \zeta)^2 > 2$ keeps the field inactive and error is at most $\rho + \zeta$.

During restoration every cell follows

$$\dot{x}_i = 5b(t)x_i(1 - x_i^2) + d_i(t). \quad (\text{B.3})$$

Within $|x_i - s_i| \leq 0.1$, the well-error upper derivative is bounded by $-1.71 \cdot 5b(t)|x_i - s_i| + \zeta$. Thus its endpoint error is at most $e_0 e^{-8.55} + \zeta$. Applying this to the worst active error gives

$$0.013334e^{-8.55} + 0.001 < 0.001003 < \rho.$$

Induction proves the exact intended decoded trace for every finite gate sequence under this bounded-disturbance contract. All work cells participate in the return; reset or reuse does not remove them from the component. These explicit bounded cells specialize established smooth-simulation methods [10] to the native gate and record contract here.

Exact decoded behavior alone does not certify every physical boundary. Disturbances must not introduce unmodeled returning couplings. The native record grammar consists of the stipulated well-sign reads; adding an analog readout changes its predictive classes. Active couplers are directed powered forces, and the equations by themselves supply no switching heat or energy rating. Preparation, holding, clock, source generation and output costs remain explicit device dependencies.

B.3 A distinct finite Gaussian refinement

For $dX_i = f_i(X, t) dt + \sigma dB_i$ with independent Brownian motions and an exact independent clock, bounded noise derivatives cannot be assumed. Inputs are nonanticipating, and the driving increments are fresh conditional on the preceding source and controller history. Choose $h = 0.002$. On each unit interval the reflection bound gives

$$\mathbb{P}\{\sup_t |\sigma(B_i(t) - B_i(0))| > h\} \leq 4e^{-h^2/(2\sigma^2)}.$$

For a scalar nonincreasing drift, a continuous forcing path bounded by h changes a solution by at most $2h$: subtract that path from the solution difference and apply a first-crossing argument at $\pm h$. This applies to writes and, by stopping inside the well, restoration. A noisy write therefore ends within $(2 + \rho)e^{-10} + 2h$ of its target. Held sources move by at most h and remain on their decoder plateaus.

For an opposite-sign SWAP, the scaled coordinates $(r/a, q)$ have noise covariance $\sigma^2 \text{diag}(9/2, 1/2)$. Rotation into the deterministic moving frame preserves its largest eigenvalue, so each coordinate martingale has quadratic variation at most $(9/2)\sigma^2$ per unit. The exponential maximal inequality and a union bound give $8e^{-h^2/(9\sigma^2)}$ for either coordinate exceeding h in absolute value. On the complementary event, the scaled perturbation is at most $\sqrt{10}\rho + \sqrt{2}h < \sqrt{3}/2 - 1$; the stopped process therefore stays in the rotating plateau. Transforming back gives endpoint well error at most $\rho + \sqrt{20/9}h$. Equal-sign pairs remain in the inactive region on the ordinary Brownian event, with error at most $\rho + h$.

Every active-unit bound leaves an error below 0.013. During restoration the noiseless path stays in its well, while the stopped $2h$ comparison stays strictly inside the 0.1 neighborhood on which the restoring drift is nonincreasing. There is consequently no stopped exit, and endpoint error is less than $0.013e^{-8.55} + 2h < 0.004003 < \rho$. Conditional on each successful preceding boundary,

independent Brownian increments permit the same bound from every point of the new well box. Union bounding the first failure over active and restoration intervals, m cells and G gates yields

$$\delta_G \leq \min\left\{1, G\left(8me^{-h^2/(2\sigma^2)} + 8e^{-h^2/(9\sigma^2)}\right)\right\}. \quad (\text{B.4})$$

Preparation failure is added separately. On the common good event the native decoded transcript equals the ideal transcript, hence their total variation distance is at most δ_G . For two arms, success differences change by at most the sum of their error bounds.

This coupling covers the stated powered bit primitives. For stochastic source ingress, its law must be exact or supplied with a separate uniform conditional kernel-error bound; an internal-gate noise estimate does not certify the external sensor. Source errors, acquisition failures and any clock failures must be added to the episode error through the same first-failure argument before applying the comparison tolerance.

Equation (B.4) is a finite-horizon approximation. The unconditional sign process need not be exactly Markov, since different voltages inside one well can have different later error probabilities. An exact SPC-2 claim for that stochastic voltage process requires its own complete continuous/native closure analysis. It cannot inherit an exact finite quotient solely from this error estimate.

B.4 Port-program stage counts and costs

Use NAND decompositions with XOR costing four gates, multiplexing four, AND two, and three-bit majority six. The inquiry circuit's two initial endorsement copies cost two gates; agreement and initialization cost eight; the masked third endorsement read costs three; priority installation costs four; each flag-update stage costs seventeen; final prediction costs fourteen. Each of the $L - 1$ later calibrations also sets a request bit, as does the final target stage. Consequently

$$2 + 8 + 3 + 4 + 17L + 14 + (L - 1) + 1 = 31 + 18L.$$

There are $L + 2$ inquiry slots of 27 ordinary windows and one final stage of 18, yielding $27L + 72$ windows. The two flags suffice because the first useful threshold cannot be reached in fewer than L calibrations, and at exactly L only the all-positive and all-negative increment histories reach it. An eight-cell scratch allocation is initialized before use in every stage. The supplied circuit tables specify each source and target; these counts do not assert minimum bank size.

For $m = 23$, each tour costs 44, proving (7.6). The $L + 1$ raw triple sites each have one joint native ingress boundary and three single-write work units, explaining the extra $2(L + 1)$ work units. An absent calibration writes constant zeros in its allocated site and retains the same cost allowance. Both arms use the same padded resources. If τ_g, E_g bound an elementary gate window and single-write work unit, and $\tau_{\text{port}}, E_{\text{port}}$ bound an acquisition interval, then

$$\begin{aligned} T &\leq (1215L + 3284)\tau_g + (L + 3)\tau_{\text{port}}, \\ E &\leq (1217L + 3286)E_g + (L + 3)E_{\text{port}} + E_{\text{aux}}. \end{aligned}$$

E_{aux} includes charged clock, program, holding and other apparatus overhead. These are conditional resource upper bounds, not thermodynamic equalities. The constant 23-cell bank does not imply constant apparatus storage: a literal unrolled program grows linearly in L , and its phase counter grows logarithmically. All source and coupler state needed for joint ingress and future laws must be modeled; a serial unobserved buffer cannot be omitted.

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